Neutron Multiplication and the Chain Reaction

Notes on Section 1.5 of E. E. Lewis,
Fundamentals of Nuclear Reactor Physics

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1 The Life of a Neutron

A U-235 nucleus in the fuel splits, and 2–3 fast neutrons fly out. These are *generation zero*. Most of them do not cause fission right away. A neutron can be absorbed at its first collision, but usually it first bounces off nuclei many times (about 18–20 times off hydrogen), slowing down, until it is finally absorbed.

Why slow down first? A slow neutron is much more likely to make U-235 fission than a fast one. A slow neutron spends more time near the nucleus, so it has more chance of being absorbed. This chance is measured by the *cross section*: the effective target area the nucleus presents to the neutron, in barns ($1\,\mathrm{b} = 10^{-24}\,\mathrm{cm}^2$, about the actual cross-sectional area of a uranium nucleus). For U-235 fission:

- fast neutron ($\sim 2\,\mathrm{MeV}$, just born): about 1 b;
- slow (thermal) neutron: about 585 b.

So a slow neutron is about 500 times more likely to cause fission. A few fast neutrons do cause fission before slowing down (in U-235 or U-238), but this adds only about 3% more fissions.

In a typical (light-water) reactor, the nuclei a neutron can bounce off are:

- **Hydrogen (H) in the water**: the main one for slowing neutrons. A proton has almost the same mass as a neutron, so the neutron loses a lot of energy per collision.
- **Oxygen (O)**: in the water and in the UO_2 fuel.
- **Uranium (U-238 and U-235)**: in the fuel.
- **Zirconium (Zr)**: in the metal tubes (cladding) that hold the fuel pellets.
- **Other materials**: fission products in the fuel, structural steel, and control rods (e.g. boron or cadmium).

Where the neutron finally ends up decides what happens next:

- If it is absorbed by U-235, that nucleus may fission and release new neutrons, the next generation.
- If it is absorbed by something else (U-238, water, cladding, a control rod), it is *captured*: it stops being a free neutron and becomes part of that nucleus, which is now one neutron heavier. It can no longer cause a fission, so for the chain reaction it is lost: it does not count toward the next generation. The heavier nucleus is often radioactive. For example:

- U-238: $n + {}^{238}_{92}\mathrm{U} \to {}^{239}_{92}\mathrm{U} + \gamma$. U-239 beta-decays to Np-239 and then to Pu-239; this is how plutonium builds up in the fuel.
- Hydrogen in water: $n + {}^{1}_{1}\mathrm{H} \to {}^{2}_{1}\mathrm{H} + \gamma$ (2.2 MeV). Deuterium is stable, so nothing radioactive is left.
- Boron in control rods: $n + {}^{10}_{5}\mathrm{B} \to {}^{7}_{3}\mathrm{Li} + \alpha$. This is why boron is such a good absorber.
- Zirconium cladding or steel: the nucleus often becomes a radioactive isotope ("activation").

- Some neutrons escape out of the core entirely.

This journey, from birth in a fission, through the bouncing, to absorption, is one **neutron lifetime**, ℓ. Then the story repeats with the new generation.

2 Neutrons per Fission

Each fission releases a whole number of neutrons, usually 2 or 3 (occasionally 0, 1, 4 or 5). Averaged over many fissions, U-235 gives about **2.4** neutrons per fission. In the examples below this is rounded to **2.5** to keep the arithmetic simple.

3 The Multiplication Factor k

The question is how many neutrons each generation produces for the next. That ratio is the multiplication factor:

$$k = \frac{\text{neutrons born in the next generation}}{\text{neutrons in this generation}}. \tag{1}$$

k counts the **new neutrons born** from fissions, not the neutrons that hit uranium. The wasted neutrons are already included, because they lower k. Think of it as counting neutrons in two rounds:

- **Round 1 (this generation):** count the neutrons, and see what each one does: wasted (absorbed by water, U-238, etc., or leaks out) or causes a U-235 fission.
- **Round 2 (next generation):** each fission releases ~ 2.5 new neutrons, so new neutrons = fissions $\times$ 2.5.
- k = round 2 $\div$ round 1.

Examples

Example 1: $k = 1$ (steady).

- Round 1: **100** neutrons.
  - Wasted: 60
  - Cause fission: 40
- Round 2: $40 \times 2.5 =$ **100** new neutrons.
- $k = 100 \div 100 =$ **1**.
- Result: $100 \to 100 \to 100 \to \cdots$ (stays the same).

Example 2: $k = 2$ **(doubling).**

- Round 1: **100** neutrons.
  - Wasted: 20
  - Cause fission: 80
- Round 2: $80 \times 2.5 = \mathbf{200}$ new neutrons.
- $k = 200 \div 100 = \mathbf{2}$.
- Result: $100 \to 200 \to 400 \to \cdots$ (doubles each generation).

Example 3: $k = 0.75$ **(dying out).** Control rods pushed in, so more neutrons are wasted.

- Round 1: **100** neutrons.
  - Wasted: 70
  - Cause fission: 30
- Round 2: $30 \times 2.5 = \mathbf{75}$ new neutrons.
- $k = 75 \div 100 = \mathbf{0.75}$.
- Result: $100 \to 75 \to 56 \to 42 \to \cdots$ (shrinks each generation; the chain reaction stops).

Example 4: $k = 1.025$ **(slowly growing).** Only one more neutron in 100 causes fission than in Example 1.

- Round 1: **100** neutrons.
  - Wasted: 59
  - Cause fission: 41
- Round 2: $41 \times 2.5 = \mathbf{102.5}$ new neutrons.
- $k = 102.5 \div 100 = \mathbf{1.025}$.
- Result: $100 \to 102.5 \to 105 \to 108 \to \cdots$ (grows 2.5% each generation). A tiny change in how many neutrons are wasted is enough to make the reactor grow.

Example 5: $k = 1$**, with the wasted neutrons broken down.**

- Round 1: **1000** neutrons.
  - Captured by U-238: 250
  - Captured by water: 150
  - Captured by cladding and control rods: 100
  - Leak out of the core: 100
  - Cause fission in U-235: 400
- Wasted in total: $250 + 150 + 100 + 100 = 600$.
- Round 2: $400 \times 2.5 = \mathbf{1000}$ new neutrons.
- $k = 1000 \div 1000 = \mathbf{1}$.
- Result: $1000 \to 1000 \to 1000 \to \cdots$ (steady). Only 40% of neutrons need to cause fission to keep the reactor steady.

4 Neutron Population in Time

Step 1: count generation by generation. Start with n_0 neutrons. By the definition of k, each generation is k times the one before it:

$$n_{\text{next}} = k \times n_{\text{this}}.$$

Apply this again and again:

$$\begin{aligned}
\text{Generation 0:} \quad & n_0 = n_0 \\
\text{Generation 1:} \quad & n_1 = k\, n_0 \\
\text{Generation 2:} \quad & n_2 = k\, n_1 = k\,(k\, n_0) = k^2\, n_0 \\
\text{Generation 3:} \quad & n_3 = k\, n_2 = k\,(k^2\, n_0) = k^3\, n_0 \\
\text{Generation 4:} \quad & n_4 = k\, n_3 = k\,(k^3\, n_0) = k^4\, n_0 \\
& \vdots \\
\text{Generation } i\text{:} \quad & n_i = k\, n_{i-1} = k\,(k^{i-1}\, n_0) = k^i\, n_0
\end{aligned}$$

Each step adds one more factor of k, so after i generations there are i factors of k:

$$n_i = k^i\, n_0.$$

Check with Example 2 ($k = 2$, $n_0 = 100$): $n_1 = 200$, $n_2 = 400$, $n_3 = 2^3 \times 100 = 800$.

Step 2: turn generations into time. Each generation lives for one neutron lifetime ℓ before the next one is born:

$$\begin{aligned}
\text{Generation 0 is born at} \quad & t = 0 \\
\text{Generation 1 is born at} \quad & t = \ell \\
\text{Generation 2 is born at} \quad & t = 2\ell \\
\text{Generation 3 is born at} \quad & t = 3\ell \\
& \vdots \\
\text{Generation } i \text{ is born at} \quad & t = i\,\ell
\end{aligned}$$

Solve the last line for i:

$$i = \frac{t}{\ell}.$$

Step 3: substitute. Put $i = t/\ell$ into $n_i = k^i n_0$. The generation number i is replaced by the time t:

$$n(t) = n_0\, k^{t/\ell}. \tag{2}$$

Here

- $n(t)$ is the number of neutrons in the reactor at time t;
- n_0 is the number of neutrons at $t = 0$, the zeroth generation (100 in the example above);
- k is the multiplication factor;
- ℓ is the neutron lifetime, so t/ℓ is the number of generations that have passed by time t.

Now we can plug in a time directly, without counting generations.

Worked examples

Every example uses the same three steps: (1) find the number of generations t/ℓ, (2) raise k to that power, (3) multiply by n_0.

Example 1: $k = 2$ (growing). $n_0 = 100$, $\ell = 0.1\,\text{ms}$. How many neutrons at $t = 0.3\,\text{ms}$?

(1) $t/\ell = 0.3\,\text{ms} \div 0.1\,\text{ms} = 3$ generations.

(2) $k^{t/\ell} = 2^3 = 8$.

(3) $n = 100 \times 8 = 800$.

Check by counting generations: $100 \to 200 \to 400 \to 800$. Same answer.

Example 2: $k = 1$ (steady). $n_0 = 100$, any time t. Since 1 to any power is 1, $n = 100 \times 1^{t/\ell} = 100$. The population never changes.

Example 3: $k = 0.9$ (dying out). $n_0 = 1000$, $\ell = 0.1\,\text{ms}$. How many neutrons at $t = 1\,\text{ms}$?

(1) $t/\ell = 1 \div 0.1 = 10$ generations.

(2) $0.9^{10} \approx 0.35$.

(3) $n \approx 1000 \times 0.35 = 350$.

After only 10 generations, about two thirds of the neutrons are gone.

Example 4: a time between generations. $k = 2$, $n_0 = 100$, $\ell = 0.1\,\text{ms}$, $t = 0.25\,\text{ms}$.

(1) $t/\ell = 2.5$ generations (not a whole number, and that is fine).

(2) $2^{2.5} \approx 5.66$.

(3) $n \approx 566$.

This lies between generation 2 (400) and generation 3 (800). The formula gives a smooth curve through the generation-by-generation steps.

Example 5: k very close to 1. $k = 1.001$, $n_0 = 100$, $\ell = 10^{-4}\,\text{s}$, $t = 1\,\text{s}$.

(1) $t/\ell = 1 \div 10^{-4} = 10{,}000$ generations.

(2) $1.001^{10{,}000} \approx 22{,}000$.

(3) $n \approx 100 \times 22{,}000 = 2.2$ million.

Even though each generation grows by only 0.1%, so many generations fit into one second that the population grows about 22,000-fold. Section 8 explains this with a simpler formula.

5 Why k Can Be Treated as a Number

1. Each generation is random. Two things change from one fission to the next:

- **Neutrons released per fission:** sometimes 2, sometimes 3, occasionally 0, 1, 4 or 5.
- **What each neutron does:** causes fission, gets captured, or leaks out.

So if we counted exactly, k would wobble a little from one generation to the next.

2. With many neutrons, the randomness averages out. It is like flipping a coin. One flip cannot be predicted, but in a million flips almost exactly half are heads. In the same way, with billions of neutrons the wobbles cancel: on average each fission gives 2.4 neutrons, and the same share of neutrons (for example 40%) causes fission every generation. So k behaves like a fixed number. $k = 1$ does not mean every neutron behaves the same way; it means that, on average, each generation exactly replaces the previous one.

3. What sets that average. The share of neutrons that cause fission is set by the reactor itself: how much U-235 there is, how much water, where the control rods are, and the size and shape of the core.

4. When k changes. k stays fixed only while the reactor stays the same. If the reactor changes, the share changes, and so does k:

- **Control rods pushed in:** more neutrons are absorbed by the rods, so k goes down (pulled out: k goes up).
- **Temperature changes:** the materials behave differently, which changes k.
- **Fuel is used up:** less U-235 and more fission products absorbing neutrons, so k goes down.

This is why the book treats k as constant only over a short time, when nothing changes. These changes are covered in later chapters.

6 Three Possible Outcomes

- $\boldsymbol{k > 1}$ **(supercritical)**: each generation is bigger, so the neutron population, and with it the fission rate and heat, grows exponentially.
- $\boldsymbol{k = 1}$ **(critical)**: each generation exactly replaces the previous one, giving a steady population and steady power. This is how a reactor normally operates.
- $\boldsymbol{k < 1}$ **(subcritical)**: each generation is smaller, and the chain reaction dies out.

7 Which k Do We Want, and When?

The reactor's power is set by how many neutrons there are (more neutrons, more fissions, more heat). k does not set the power itself; it only says whether the neutron number is rising, steady or falling. So each case has a job:

$k = 1$: normal operation. A power plant needs steady heat to make steady electricity, so it runs at $k = 1$ almost all the time. The power stays at whatever level the neutron population is at.

k slightly above 1: only briefly, to raise power. At start-up, or when more power is needed, operators pull the control rods out a little. k goes slightly above 1, the neutron population and power rise, and then they bring k back to 1 so the power holds at the new, higher level.

Why we don't want a large or long-lasting $k > 1$: the population keeps growing exponentially, so the power keeps rising until the fuel overheats and can melt. The 1986 Chernobyl accident was

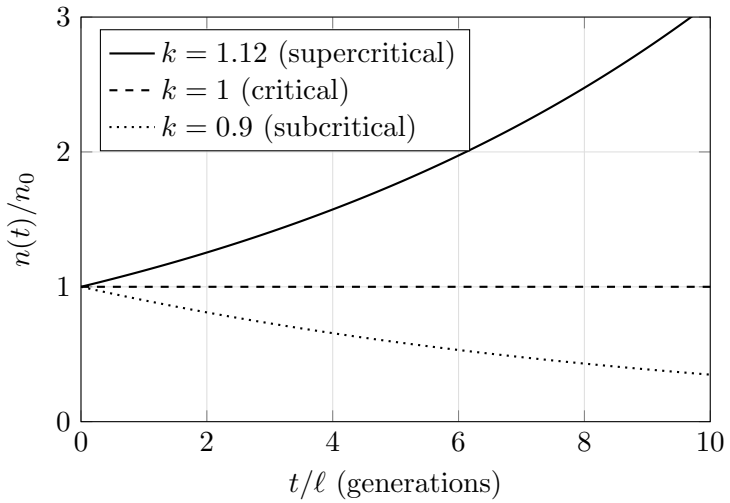


Figure 1: Neutron population versus time for the three cases (compare Fig. 1.3 in Lewis).

such a runaway power rise. That is why $k - 1$ is always kept tiny, well below about 0.0065 (the fraction of delayed neutrons for U-235, covered later), so any rise is slow enough for operators and control systems to follow.

k below 1: to lower power or shut down. Inserting control rods pushes k below 1 and the power falls. An emergency shutdown (a "scram") drops all the rods in at once. During refueling and maintenance the reactor is kept well below $k = 1$; this safety cushion is called the *shutdown margin*.

Why we don't want $k < 1$ while producing power: the chain reaction dies out, and so does the heat and electricity.

In short. $k = 1$ holds the power steady, $k > 1$ raises it, and $k < 1$ lowers it. The skill is in keeping k very close to 1 and moving it only slightly and briefly.

8 Reactor vs. Atomic Bomb

Both use a chain reaction, so it is natural to ask: if k rises above 1 in a reactor, does it become a bomb? **No.** They are built to behave in completely different ways.

	Reactor	**Bomb**
Goal	steady heat for years	all energy in an instant
k	kept at 1 (at most slightly above, e.g. 1.001)	about 2
Fuel	3–5% U-235, rest U-238	over 90% U-235, or Pu
Neutrons	slowed down by water	fast, nothing slows them
Time per generation	$\sim 10^{-4}$ s, plus delayed neutrons	$\sim 10^{-8}$ s
Control	control rods, operators	none

Why reactor fuel cannot explode like a bomb.

- Most of the fuel is U-238, which captures many neutrons instead of fissioning. This keeps k low.
- The chain reaction in a reactor relies on *slow* neutrons. Slowing down takes time, so each generation is slow.
- A small fraction of neutrons are released late (delayed neutrons), which makes any rise in power slow enough to control.

What happens if k rises too much in a reactor. The power climbs until the fuel overheats. This can melt the fuel or cause a steam explosion, as at Chernobyl in 1986. It is serious, but it is *not* a nuclear explosion.

In short. A reactor is a slow, controlled chain reaction with $k \approx 1$. A bomb is a fast, uncontrolled one with $k \approx 2$.

9 The Case $k \approx 1$

Even a tiny difference from 1 matters. Because ℓ is very short (well under a millisecond), there are thousands of generations every second. Using $x = \exp[\ln x]$, Eq. (2) can be written

$$n(t) = n_0 \exp\left[\frac{t}{\ell} \ln k\right]. \tag{3}$$

When k is close to 1, $|k-1| \ll 1$, we can use $\ln k \approx k - 1$, so

$$n(t) \approx n_0 \exp\left[\frac{(k-1)\, t}{\ell}\right], \tag{4}$$

and the population changes quickly.

That is why operators control k so carefully, using control rods (which absorb neutrons) and the fuel design. The rest of the book is about how to calculate k and how to keep it at exactly 1.